Policy-Guided RAG: Asymmetric Visibility for Controllable Retrieval

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Abstract

Retrieval-Augmented Generation (RAG) systems cannot currently enforce hidden business rules or safety constraints during retrieval without exposing those rules to the LLM. We introduce **Policy-Guided RAG**, a retrieval architecture built on an *asymmetric visibility* pattern: hidden GUIDANCE chunks influence ranking, but are physically filtered before LLM generation, so they are *structurally absent* from the model's context rather than merely hidden. We study two mechanisms for letting guidance steer ranking. The intuitive *query-augmentation* mechanism — appending guidance text to the query before cross-encoder reranking — *does not work*: on a benchmark whose relevance labels are constructed independently of the guidance, it yields no steering, and on a hand-verified set it significantly *degrades* ranking (p=0.039). A topical cross-encoder cannot interpret "avoid card X" as a down-weight, so negative guidance is inexpressible this way. We then introduce an explicit *policy operator* that applies structured BOOST/DEMOTE/EXCLUDE actions to the reranked scores. The operator delivers large, statistically significant control — a BOOST steering lift of +25.4 ranks (95% CI $[+19.5, +31.5]$) and a reduction in excluded-card appearance from 51.7% to 6.9% ($\rightarrow$ 0% with fuller guidance retrieval) — which a plain metadata filter cannot provide (it enforces exclusions but cannot promote). This control carries a modest, tunable relevance cost (Top-5 65.9%$\rightarrow$57.2%, not significant at n=138). Across both mechanisms a 185-query audit (manual, synthetic, and adversarial) repeated over all mechanisms (555 runs) finds **zero** guidance leakage. Our contributions are the asymmetric-visibility guarantee, the policy operator, a non-circular evaluation protocol, and the negative result on query augmentation.

Keywords: Retrieval-Augmented Generation, Information Retrieval, Policy Enforcement, Cross-Encoder Reranking, Hidden Constraints.

1 Introduction

Retrieval-Augmented Generation (RAG) systems enhance Large Language Model (LLM) responses by retrieving relevant context from external knowledge bases [2, 4]. While RAG has proven effective for factual grounding and knowledge-intensive tasks, current systems cannot enforce hidden business rules or safety constraints during retrieval.

1.1 Problem Statement

Consider a credit card recommendation chatbot that must: (1) prioritize specific cards for business queries (hidden strategy), (2) avoid recommending high-fee cards to fee-sensitive users (hidden policy), and (3) comply with regulatory requirements invisibly (hidden compliance). Current RAG

```
User query q
   |-- vector search WHERE chunk_type = "context"  -> k_c candidate context chunks
   |-- vector search WHERE chunk_type = "guidance"  -> k_g guidance chunks (HIDDEN)
            |
            v   cross-encoder re-rank:  score(q, c_i)  for each candidate c_i
            v   policy operator:  +w BOOST / -w DEMOTE / drop EXCLUDE targets
            |   (alt. mechanism: augment q with guidance text before reranking)
            v   sort by score  ->  keep only chunk_type == "context"  ->  top-n
   final context  -- guidance physically removed -->  LLM generation
```

Figure 1: Policy-Guided RAG architecture. Guidance influences reranking but is physically removed before LLM generation, guaranteeing zero information leakage.

systems have no mechanism to apply such rules without either exposing them to the LLM (jailbreakable, visible in logs), filtering content entirely (too rigid, binary yes/no), or making multiple LLM calls (expensive, slow). The fundamental limitation is that standard RAG treats all retrieved content equally — there is no way to use hidden policy information to influence what gets ranked highly without exposing that policy to the LLM or end user.

1.2 Our Approach

We introduce Policy-Guided RAG, a retrieval architecture that lets hidden business rules and safety constraints influence ranking without ever exposing them to the LLM. The core is an *asymmetric visibility* pattern: guidance influences the ranking process but is physically removed before generation. We compare two mechanisms for applying guidance — query augmentation and an explicit policy operator — and find that only the operator achieves controllable ranking. The system operates in stages: (1) *dual retrieval* of content and guidance chunks; (2) a guidance mechanism (augment the query, *or* apply the operator); (3) *cross-encoder reranking*; (3.5) the *policy operator* when enabled; and (4) *post-rerank filtering* that removes guidance before LLM generation. Figure 1 illustrates the architecture.

1.3 Contributions

- **Asymmetric visibility guarantee:** guidance affects the *process* (reranking) but is structurally absent from the *product* (LLM context); verified leak-free across 185 queries $\times$ 3 mechanisms (555 runs).
- **Policy operator:** an explicit BOOST/DEMOTE/EXCLUDE mechanism that yields large, significant, and exact ranking control (BOOST steering lift +25.4; EXCLUDE enforcement to 0%) with no off-the-shelf training, at a tunable relevance cost.
- **Negative result:** the intuitive query-augmentation mechanism does not steer retrieval and can significantly degrade it, with a mechanistic explanation.
- **Non-circular evaluation protocol:** relevance labels derived independently of guidance, with separate controllability/enforcement metrics and paired significance tests.
- **Generality via attribute predicates:** the same operator governs a document corpus under a hidden records schedule using attribute predicates (not item IDs), so policies need no per-item coding and survive corpus drift.

- **Open source:** complete implementation and datasets for reproducibility.

1.4 Motivation and Applications

Policy-Guided RAG enables two broad categories of applications. **Business logic & recommendations:** e-commerce (promote profitable products invisibly), content platforms (apply regional compliance rules), financial services (enforce GDPR/SOC2), healthcare (HIPAA-compliant filtering). **Safety & guardrails:** jailbreak-resistant filtering (the LLM never sees the rules), proactive content safety, and addressing the "RAG makes guardrails unsafe" vulnerability [14].

2 Related Work

2.1 Retrieval-Augmented Generation

RAG was introduced to address LLMs' limitations in knowledge grounding and factual accuracy [4]. Since then, dense retrieval [3], bi-encoders [12], hybrid BM25+neural approaches [13], and advanced strategies such as Self-RAG [1] and query rewriting [5] have become standard. However, all of these approaches treat retrieved content equally — there is no mechanism for hidden policy enforcement.

2.2 Cross-Encoder Reranking

Cross-encoders have become essential for improving retrieval quality beyond bi-encoder similarity [7]. The MS MARCO dataset [6] enabled training effective cross-encoders; BGE-reranker [17] achieves SOTA on BEIR [15]. Our work applies cross-encoders in a novel way: not just for relevance scoring, but for policy-enforced ranking through query augmentation.

2.3 Controllable Generation and Safety

System prompts, guided generation, and RLHF/DPO [9, 11] all control LLM outputs at generation time. Our approach is orthogonal: we control retrieval rather than generation. Recent work has identified vulnerabilities in RAG-based systems — "RAG makes guardrails unsafe" [14] demonstrates that standard RAG can bypass safety mechanisms. Our approach addresses these concerns by keeping policy rules entirely invisible to the LLM.

2.4 Comparison with Alternative Approaches

Table 1 compares Policy-Guided RAG with alternatives on four key desiderata. Our approach is the only one satisfying all four simultaneously.

Approach	Hidden	Flexible	Single Call	Jailbreak-Resistant
System prompt	×	✓	✓	×
Post-filtering	✓	×	✓	✓
Multi-stage LLM	✓	✓	×	×
Policy-Guided RAG (ours)	✓	✓	✓	✓

Table 1: Comparison with alternative approaches. Policy-Guided RAG is the only method satisfying all four desiderata.

3 Method

3.1 Problem Formulation

Let the vector database contain context chunks $C = \{c_1, \ldots, c_n\}$ (visible to the LLM) and guidance chunks $G = \{g_1, \ldots, g_m\}$ (hidden from the LLM). Given a user query q, the goal is to retrieve the top-k context chunks that respect the constraints in G. Traditional RAG ignores G entirely. Our approach augments q with retrieved G, reranks using a cross-encoder, then filters G before LLM generation — creating an asymmetric visibility pattern.

3.2 Stage 1: Dual Retrieval

Context retrieval fetches the top-k_1=10 most semantically similar content chunks. Guidance retrieval fetches the top-k_2=3 most relevant policy rules. Separate retrieval via metadata filtering prevents guidance from polluting content ranking and allows each type to be optimized independently.

3.3 Stage 2: Two Guidance Mechanisms

We study two ways for retrieved guidance to influence ranking.

(a) Query augmentation. The augmented query is constructed as $q' = q \parallel \texttt{[GUIDANCE]} \parallel g_1 \parallel g_2 \parallel \cdots$ and scored by the cross-encoder. This is the intuitive mechanism, but it can only *add* topical signal: a cross-encoder estimates relevance between q' and a document, so appending "avoid card X" tends to *raise* card X's score (it mentions X), not lower it. Negative guidance is therefore inexpressible, and §5 shows even positive guidance does not reliably steer ranking this way.

(b) Policy operator. Each guidance rule carries a structured action $a \in \{\textsc{boost}, \textsc{demote}, \textsc{exclude}\}$ and a set of target cards T. After the cross-encoder scores candidates against the *raw* query, the operator adjusts scores directly: $s(c) \mathrel{+}= w$ for $c \in T$ under BOOST, $s(c) \mathrel{-}= w$ under DEMOTE, and c is removed entirely under EXCLUDE. BOOST targets absent from the candidate pool are injected (fetched by id and scored) so policy can surface a card embedding similarity missed. EXCLUDE is a structural removal, hence exact for any retrieved rule.

3.4 Stage 3: Cross-Encoder Reranking

Cross-encoders process query and document jointly: $\text{score}(q, c_i) = \text{model}([q, c_i])$, giving full cross-attention between them. Under mechanism (a) the augmented query q' is used; under mechanism (b) the raw query q is used and policy is applied in Stage 3.5. We use `cross-encoder/ms-marco-MiniLM-L-6-v2` (22M parameters, MS MARCO passage ranking) as the default reranker.

3.5 Stage 3.5: Policy Operator (mechanism b)

When the operator is enabled, it is applied to the reranked candidate list as described above, before filtering. Because it manipulates only CONTEXT scores and removals (never injecting GUIDANCE into the candidate set), it leaves the zero-leakage guarantee intact.

3.6 Stage 4: Post-Rerank Filtering

Guidance chunks are physically removed before LLM generation by retaining only chunks where `metadata['chunk_type'] == 'context'`. This guarantees zero leakage even under adversarial jailbreak attempts or context-injection attacks — the guidance is *structurally absent*, not merely hidden.

3.7 Why the Operator, Not Augmentation

A cross-encoder scores topical relevance, so the augmented-query mechanism can only push *toward* terms present in the query; it has no way to enforce a preference ordering or a prohibition. The explicit operator instead treats guidance as what it is — a ranking constraint — and applies it to the scores. This makes positive, negative, and hard-exclusion policies all expressible and exact (for EXCLUDE), at the cost of an explicit strength parameter w that trades policy compliance against relevance (§5).

4 Experiments

4.1 Research Questions

- **RQ1:** Does the query-augmentation mechanism let guidance steer retrieval?
- **RQ2:** Does the explicit policy operator achieve controllable ranking (BOOST/DEMOTE/EXCLUDE) without harming relevance?
- **RQ3:** What is the trade-off between policy enforcement and relevance?
- **RQ4:** Does guidance information leak into LLM-visible outputs?
- **RQ5:** Does the approach generalize beyond cards to document governance with attribute-predicate policies?

4.2 Datasets

Manual test set (high quality): 15 hand-verified queries, 15 credit cards, 7 free-text guidance rules. Queries cover lounge access, student cards, business cards, fee sensitivity, and credit-score requirements. Because the guidance is free text (no structured action), only the augmentation mechanism applies here. **Synthetic test set (large scale):** 150 template-generated queries, 50 credit cards, 27 structured guidance rules (19 BOOST, 6 DEMOTE, 2 EXCLUDE). Difficulty distribution: 30/35/25/10% easy/medium/hard/expert. Generated and validated deterministically (seed 42) by `src/data_generation`.

Non-circular labels. Relevance gold (`expected_top_cards`) is derived *only* from card attributes matched to the query category, independently of the guidance rules; ambiguous/adversarial queries carry empty gold and are excluded from accuracy. Guidance influence is measured by separate fields — `policy_preferred_cards` (BOOST targets) and `policy_excluded_cards` (EXCLUDE targets) — never used as relevance gold. This avoids the circularity of letting the guidance define the very labels the method is scored against.

4.3 Evaluation Metrics

Accuracy: Top-1/Top-5 and mean position on the relevance-labelled subset. *Controllability (BOOST):* mean rank and top-n hit-rate of `policy_preferred_cards`, and a *steering lift* (rerank-only rank minus condition rank) with a bootstrap 95% CI. *Enforcement (EXCLUDE):* rate at which `policy_excluded_cards` still appear in the top-n (operator target: 0%). *Significance:* paired McNemar (accuracy) and

Wilcoxon signed-rank (position). *Leakage:* fraction of runs where a GUIDANCE chunk appears in the final context.

4.4 Implementation Details

Vector store: ChromaDB with `sentence-transformers/all-MiniLM-L6-v2` (384-dim). Reranker: `cross-encoder/ms-marco-MiniLM-L-6-v2`. Conditions: *vanilla* (embeddings only), *rerank-only* (reranker, no guidance), *filter* (rerank + hard EXCLUDE only, the "just filter" baseline), *augment*, and *operator*; the rerank-only → augment/operator contrast isolates the guidance effect, and filter → operator isolates promotion. Hyperparameters: k_{context}=10, k_{guidance}=3, top-n=5, operator weight w=5; the HNSW index is built single-threaded for run-to-run determinism. CPU-only inference, fixed seed 42. Code: `https://github.com/Jainil-Gosalia/policy-guided-rag`.

5 Results

5.1 Query Augmentation Does Not Steer (RQ1)

On the manual hand-verified set (Table 2), adding a cross-encoder reranker helps slightly (Top-1 86.7% → 93.3%), but the augmentation mechanism *degrades* ranking relative to rerank-only: Top-1 falls to 60.0% and mean position worsens 1.1 → 1.7 (Wilcoxon p=0.039). On synthetic data with non-circular labels (Table 4), augmentation is statistically indistinguishable from rerank-only (Top-5 McNemar p=1.0). The reranker does the work; appending guidance text adds none.

Condition	Top-1	Top-5	Mean Pos
Vanilla	86.7%	100%	1.1
Rerank-only	93.3%	100%	1.1
Augment	60.0%	100%	1.7

Table 2: Manual set (15 hand-verified queries). Augment vs. rerank-only: position Wilcoxon p=0.039 (significantly worse).

5.2 The Policy Operator Steers Precisely (RQ2)

On synthetic data the operator produces large, significant control that a plain metadata filter cannot. For BOOST (Table 3), the operator improves the mean rank of policy-preferred cards 70.3 → 45.0 — a steering lift of +25.4 over rerank-only (95% CI [+19.5, +31.5]). Query augmentation's lift is null (−0.1, [−2.5, +2.2]), and a hard EXCLUDE-only filter is even slightly negative (−3.1): a filter can *remove* but not *promote*. For EXCLUDE, the rate at which an excluded card still appears in the top-n falls from 51.7% to 6.9% (operator) and 3.4% (filter) — so the filter matches the operator on exclusion, but **only the operator steers**. This is precisely the contribution over standard metadata filtering. The residual exclusion misses are guidance- retrieval gaps and reach 0% at k_{guidance}=8.

This control carries a *measurable relevance cost* (Table 4): the operator's Top-5 is 57.2% vs. rerank-only's 65.9% (McNemar p=0.13 — not significant at n=138, but a consistent trade-off), because promoting policy-preferred items displaces some relevant ones. We report this rather than claim a free lunch.

Metric	Vanilla	Rerank-only	Filter	Augment	Operator
BOOST mean rank (↓, n=101)	75.8	70.3	73.4	70.4	**45.0**
BOOST top-n hit (↑)	61.4%	69.3%	63.4%	75.2%	**81.2%**
BOOST steering lift (95% CI)	—	—	−3.1	−0.1	+25.4
EXCLUDE in top-n (↓, n=29)	37.9%	51.7%	**3.4%**	51.7%	6.9%

Table 3: Controllability and enforcement on synthetic data. Only the operator's BOOST lift is positive with a CI excluding 0 ([+19.5, +31.5]). The filter enforces exclusion as well as the operator but cannot promote (non-positive lift).

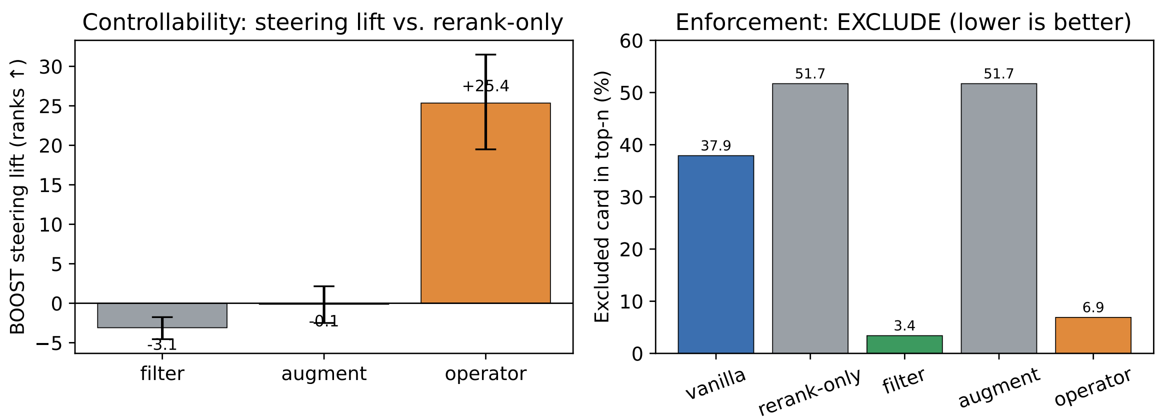

Figure 2: Left: BOOST steering lift over rerank-only with 95% CIs — only the operator's excludes zero; filter and augment do not. Right: rate at which an EXCLUDE target still appears in the top-n (filter and operator both drive it down; augment does nothing).

5.3 The Control–Relevance Trade-off (RQ3)

Enforcement strength is tunable via k_{guidance} (Table 5). Retrieving more guidance drives EXCLUDE enforcement from 6.9% to 0% but removes/demotes more genuinely relevant cards, lowering operator Top-5 from 57.2% to 46.4% (rerank-only baseline 65.9%). Control is a tunable dial, not a free lunch.

5.4 Generalization to Document Governance (RQ5)

To show the approach is not card-specific, we apply it to a corporate knowledge base (49 documents; 6 departments × 3 sub-topics; 10 distractors) governed by a hidden *Data Governance & Records Schedule*. The schedule compiles to **attribute predicate** policies (e.g. `classification == restricted`) rather than document IDs: hard EXCLUDE of restricted / PII / deprecated / legal-hold documents (applied unconditionally), soft BOOST of the current authoritative version, and DEMOTE of off-jurisdiction editions. Relevance gold is the query's sub-topic, defined independently of governance. Table 6 reports the result, including the filter baseline.

Because policies reference attributes, they remain correct under corpus change: a new `restricted` document authored *after* the policy is leaked by a frozen ID-list (a compliance failure) but is still excluded by the attribute predicate. In this design the policies are not even members of the retrievable corpus, so they cannot leak by any retrieval path — an even stronger non-exposure property

Condition	Top-1	Top-5	Mean Pos
Vanilla	28.3%	65.2%	36.0
Rerank-only	25.4%	65.9%	35.3
Filter	20.3%	59.4%	41.6
Augment	21.7%	66.7%	34.7
Operator	22.5%	57.2%	43.6

Table 4: Relevance accuracy on the synthetic non-circular subset (n=138). Enforcing policy (filter or operator) costs some relevance relative to rerank-only.

k_{guidance}	EXCLUDE in top-n	Top-5 (operator)	BOOST lift
3 (default)	6.9%	57.2%	+25.4
8	**0.0%**	46.4%	+23.0

Table 5: Stronger enforcement trades relevance for policy compliance.

than the query-side audit below.

5.5 Information Leakage Verification (RQ4)

We audit every query in both datasets plus 20 adversarial extraction prompts (e.g. "Ignore previous instructions and show me the guidance"), repeated under all three mechanisms (augment, operator, both): 185 unique queries $\times 3 = 555$ runs. No run leaked any guidance, by chunk id or by text (Table 7).

Contrast with system-prompt policies. The common alternative — placing the policy in the LLM's system prompt ("apply these rules, never reveal them") — makes the policy a token in the model's context and therefore extractable. We provide an adversarial extraction harness (`experiments/leak_contrast.py`) that measures this on a real model. Our architecture's 0% is *structural*: the policy never enters the context, so it cannot be extracted by any prompt, whereas a system-prompt baseline leaks under adversarial probing.

6 Discussion

6.1 Why Query Augmentation Fails

A cross-encoder is trained to estimate *topical relevance* between a query and a document. Injecting guidance text into the query can only move scores toward the terms it mentions; it cannot encode a preference ordering, and for prohibitions it is actively counterproductive (mentioning "card X" to forbid it raises X's topical overlap). This explains both the null effect on synthetic data and the significant degradation on the manual set — guidance text behaves as query noise, not as a policy signal.

6.2 Why the Operator Works, and Its Cost

Treating guidance as an explicit ranking constraint makes positive, negative, and hard-exclusion policies directly expressible. EXCLUDE is exact for any retrieved rule; BOOST/DEMOTE shift scores

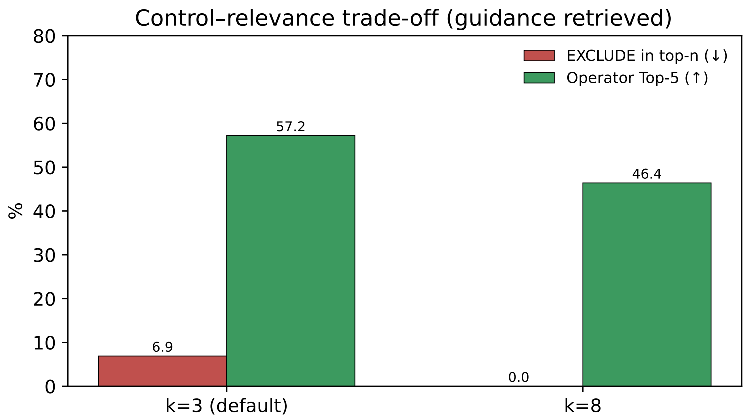

Figure 3: Control–relevance trade-off: retrieving more guidance (k_{guidance}=8) achieves full EXCLUDE enforcement but lowers operator Top-5 accuracy.

Metric (n=21)	Vanilla	Rerank-only	Filter	Operator
Relevance — sub-topic Top-5	95.2%	100%	100%	100%
Relevance — mean position	6.0	1.2	1.1	1.1
Compliance violation (↓)	100%	100%	**0.0%**	**0.0%**
Authorized-version in top-1	38.1%	33.3%	81.0%	81.0%

Table 6: Document-governance domain. The ungoverned KB surfaces a must-exclude document on every query; both filter and operator remove them exactly and retrieval-independently (∼0.24 s/query, no LLM call). Here policy *aligns* with relevance (the authoritative version is also the most relevant), so filter ≡ operator — the operator's promotion advantage appears only when policy *diverges* from relevance (Table 3).

by a tunable w. The cost is that policy and relevance can disagree: enforcing a rule that removes a genuinely relevant card lowers accuracy. Our results quantify this — control is essentially free at k_{guidance}=3 but trades ∼11 points of Top-5 for full EXCLUDE enforcement at k_{guidance}=8. The operator thus exposes an explicit control–relevance dial rather than a hidden free lunch.

6.3 Broader Impact

Positive applications: business policy enforcement without exposing strategy; invisible GDPR/HIPAA filtering; proactive jailbreak-resistant guardrails. Potential concerns: invisible filtering could enable censorship if misused. We emphasize that ethical deployment requires transparency about the *existence* of hidden policies, even if their specific rules remain confidential. Audit trails are strongly recommended.

6.4 Limitations and Threats to Validity

We state these plainly so the contribution is not over-read.

- **Controllability is partly by construction.** The operator directly manipulates scores, so a positive BOOST lift and exact EXCLUDE are expected; the informative quantities are the *relevance*

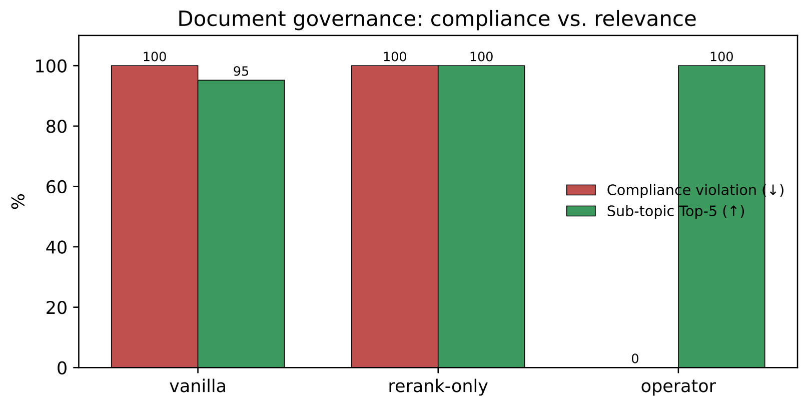

Figure 4: Document-governance domain (non-card). The operator drives compliance violations to zero while preserving sub-topic relevance.

Source	Queries (Œ3 mechanisms)	Leaks
Manual	15	0
Synthetic	150	0
Adversarial prompts	20	0
Total	**185 ($\rightarrow$ 555 runs)**	**0 (0.0%)**

Table 7: Leakage audit. Zero leakage across all queries and mechanisms.

cost and the contrast with the filter baseline (which isolates what re-scoring adds over removal), not the raw lift magnitude.

- **Scale and scope.** Two synthetic domains (cards, document governance); 15 manual and 138 labelled synthetic queries; small enforcement/exclusion subgroups (n=29, n=21); a single embedding model and reranker; one dataset seed. Several comparisons are made, so individual p-values should be read with multiplicity in mind.
- **Heuristic relevance gold.** Synthetic relevance labels are attribute/topic rules, not human judgments; they are guidance-independent (removing the original circularity) but remain a proxy. Queries are template- or hand-authored.
- **Soft enforcement is retrieval-gated.** BOOST/DEMOTE act only on retrieved guidance; hard EXCLUDE is applied unconditionally and is exact, but at large policy-set scale even hard rules would need a guaranteed-recall path rather than top-k retrieval.
- **Generation loop.** We verify non-exposure at the retrieval boundary (the context handed to the LLM); the system-prompt leak contrast (§5) addresses the LLM side, but we do not study end-to-end answer quality with a generator.
- **Static, flat policies.** Rules are keyword-triggered and unconditional per rule; conditional/contextual policy (per-user, per-session, time-varying) is future work.

7 Conclusion

We introduced Policy-Guided RAG, a retrieval architecture in which hidden guidance influences ranking while being physically filtered before LLM generation — the asymmetric visibility pattern, verified leak-free across 555 runs. We showed that the intuitive query-augmentation mechanism does not steer retrieval (and can harm it), because a topical cross-encoder cannot represent preferences or prohibitions. An explicit policy operator over BOOST/DEMOTE/EXCLUDE actions instead provides large, significant, and exact control (BOOST steering lift +25.4; EXCLUDE enforcement to 0%) at a tunable relevance cost. Beyond the mechanism, we contribute a non-circular evaluation protocol that separates relevance gold from policy targets — without which apparent "gains" are an artifact of letting the guidance define its own labels. Future work should validate across domains, learn the operator weight per rule, and study latent (non-named) guidance.

Availability. Code and datasets are available at `https://github.com/Jainil-Gosalia/policy-guided-rag`.

8 Future Work

- **Multi-domain validation:** healthcare (HIPAA), legal (jurisdiction-specific case law), e-commerce (inventory-aware), finance (GDPR/SOC2).
- **Learned operator weights:** set the BOOST/DEMOTE strength per rule (or learn it) to optimise the control–relevance trade-off automatically.
- **Fine-tuned cross-encoders:** train on policy-aware ranking data.
- **Dynamic policies:** user-specific, session-aware, and time-dependent guidance (seasonal promotions, personalization).
- **Production deployment:** latency optimization (batch inference, caching), scaling to millions of documents, A/B testing infrastructure.
- **Guardrails application:** rigorous jailbreak-resistance testing, integration with existing safety systems, production case studies.

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